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Including the Hybrids: The Thread Arrangement as a Lens for Manifesting Textile Designers' Agency

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Current textile classification systems often divide fabrics into mutually exclusive categories such as weaves, knits, and nonwovens. While useful for distinguishing basic structural differences between textiles, such taxonomies frequently fail to account for hybrid structures and may inadvertently discourage research into them. This paper proposes the notion of the "thread arrangement" as a conceptual lens to nuance such rigid classifications and to manifest the creative agency of the textile designer. By synthesising insights from textile design, creativity, and classification theory with potential-focused perspectives from textile archaeology, the term is developed as a meta-category that emphasizes the duality of the textile as both a finished result and its previous deliberate act of arranging. Through the reflective analysis of terminology developed in the author's practice-based doctoral research and different types of textile hybrids, the paper demonstrates how this lens makes structural transgressions visible and describeable. The findings suggest that shifting the focus from categorical definitions to structural potentials can allow for a more inclusive understanding of textile hybrids. The thread arrangement framework provides a space for reflective discovery, acknowledges textiles as open-ended systems of arranged threads, and reinforces the textile designer's role as their hands-on creator.

Keywords: textile design; thread arrangement; agency; hybridity.

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1. Introduction

Textiles are inherently human-made and the outcomes of a deliberate design process determining their material and structural composition (Albers, 1971, 2017). Their aesthetic and functional properties such as look, feel, density, elasticity and drape are determined by the textile designer's agency in selecting threads with a specific material composition and deciding on their structural arrangement.

Reference works for textile-related professions tend to divide textiles into interlacements (e.g. weaves or braids), interloopings (e.g. knits or crochets) and haphazard fibre alignments (e.g. nonwovens or felts) (cf. e.g. Emery, 1966; Fashionary, 2020; Sarkar et al., 2023; Thompson, 2014). While such classifications are useful to understand the principal differences between textile constructions, they make hybridisations of fabric structures that belong to several categories difficult to classify. At the same time, research into textiles tends to focus on their material composition, while alternative setups of the textile structure beyond these established categories remain comparatively underexplored. However, it is precisely this “inventive use of elements of construction” (Albers, 2017, p. 43) that could make significant contributions to the discovery of textiles with new aesthetic and functional potentials (Drews, forthcoming).

Consequently, a more or less siloed division of textiles into weaves, knits and nonwovens may fall short of encompassing the outcomes of creative explorations of alternative structural compositions or even discourage research into their hybridisation in the first place. This paper therefore proposes the notion of the "thread arrangement" as a conceptual lens for textile practitioners that could accompany the study of current classifications, thus potentially increasing the visibility of textiles transgressing habitual norms and fostering additional research into alternative ways of structuring threads within a textile. This lens is developed on the basis of terminology used to describe my practice-based doctoral research, and of examples of weave/knit hybrids developed in its course. It is further embedded into a larger interdisciplinary theoretical framework that describes textile design as a creative, hands-on research discipline, critically engages with the boundaries of classification, compares different conceptual angles underlying existing textile classifications, and presents an example for a potential-focused textile meta-category from the field of textile archaeology.

2. Theoretical framework: Unbounding textile design

Three perspectives inform the theoretical framework for the notion of the thread arrangement. The first acknowledges the agency of textile designers to decide about the precise structural and material makeup of a textile during their creative process. The second outlines potential challenges related to classification and categorisation more generally, while the third focuses specifically on potential- and practice-focused categorisations of textiles.

2.1. Textile design: Creative discovery beyond the semi-finished good

In the existing framework of production, textiles are frequently considered semi-finished goods destined to be processed further into e.g. garments, furniture coverings, or other textile-based artefacts. This vision entails a separation of tasks between textile designers responsible for determining the structural arrangement and material composition of the threads of the textile, and

disciplines like garment or product design responsible for transforming the textile into finished products (Albers, 1971, 2017; Drews, 2019, forthcoming; McQuillan, 2020; Vuletich, 2015). “A dress cannot exist without a textile” (Igoe, 2021, p. 172), yet this separation between the production of the textile and its further transformation can obscure the essential role the textile plays in the final artefact (Drews, forthcoming; Igoe, 2021; Vuletich, 2015). In addition, it leads to the frequent consideration of textiles as base materials whose production process is terminated and whose material and structural composition can therefore no longer be modified (Drews, forthcoming). However, as a growing body of research demonstrates, it is precisely their position at a relatively early stage in the production chain that provides textile designers with the agency of designing textiles as more than semi-finished goods, e.g. by integrating the aspect of the final form the textile is to assume directly into their design process (cf. e.g. Buso et al., 2022; Drews, forthcoming; Drews et al., 2023; McQuillan, 2020; Piper, 2019; Rosato et al., 2023; Voorwinden et al., 2025; Voorwinden & McQuillan, 2025).

Such different designs are often fostered by the hands-on interaction with materials and fabrication tools characteristic of experimental textile design processes reconciling design and craft, which generate experiential and embodied knowledge (Albers, 1971, 2017; Drews, forthcoming; Igoe, 2021; Nimkulrat, 2012; Pajaczkowska, 2016; Roxburgh, 2021). Textile designers often work in close proximity to the materials they use (threads, predominantly), and obtain detailed knowledge of manual production processes during their education. According to Barati and Karana (2019), it is precisely through "skillful, embodied, distributed acts of tinkering, experimenting, and making” (p. 118) that novel affordances of materials can be discovered. Their Materials Potential Framework builds on Glăveanu’s (2012) model of creativity, according to which “creativity becomes manifest in craftwork through acts of spontaneous discovery, invention of techniques and transgression of norms” (p. 205), thus revisiting the boundaries of what is considered normal in an established system. Lehmann (2012) describes a similar dynamic by stating that the generation of new knowledge of the potentials of a craft necessitates a "deliberate departure" (p. 151) from established rules and traditions.

In the specific context of textile design, it can therefore be argued that their profound technical and experiential knowledge and close proximity to the material during their creative process makes textile designers apt at discovering new possibilities for arranging threads if they are willing to deviate from the way things are usually done. This creative agency includes e.g. the exploration of hybrid textile structures that are conventionally presented as distinct entities, which are a focus of this paper.

2.2. Boundaries of classification and categorisation

Classification and categorisation are inextricably linked to scientific enquiry and world-ordering (Wilkins & Ebach, 2014), as well as essential methods for material selection in design (Ashby & Johnson, 2002). As Jacob (2004) underlines, both terms are often employed synonymously yet require nuancing: Both are “mechanisms for establishing order through the grouping of related phenomena” (p. 527), but while a classification establishes a rigid organisation in which each entity may only belong to one class, in “a categorization system, […] membership of an entity in any one category is nonbinding and does not prohibit membership in any other category” (p. 529). Therefore, the adherence to one or several categories may change over time or within different contexts, and “category boundaries are not simply fuzzy but […] mutable and potentially fluid” (ibid.). In a similar

vein, though not distinguishing as sharply between classification and categorisation, Bowker and Star (1999) underline the arbitrariness of setting up boundaries between classes and/or categories. They define "a classification [as] a spatial, temporal, or spatio-temporal segmentation of the world" (p. 10), which presents itself as a unique, complete system containing mutually exclusive categories. However, these segmentations are usually designed for specific ends and within specific contexts and almost always generate "cases that do not fit" (p. 9), which means that they usually cannot depict complex, interrelated realities in their entirety. To account for these circumstances, they formulate methodological tools for the design of classification systems. These include an "incorporation of ambiguity - leaving certain terms open for multiple definitions across different social worlds" (p. 324), the system's "ability to [...] change with changing natural, organizational, and political imperatives" (p. 325), as well as a "sensitiv[ity] to exclusions" (ibid.).

These considerations underline that, not unlike textiles, boundaries drawn between classes and categories are human-made and thus informed by the specific angles of reflection and requirements of their context of creation.

2.3. Practice- and potential-focused classifications and categorisations of textiles

Considerations of context-dependent distinction and exclusion such as the ones presented above can also be applied to classifications and categorisations of textiles. During their education, many textile professionals learn about the possibilities for arranging threads through standard reference works, which often distinguish between textiles according to their structural differences. Many of these works dedicate separate chapters to interlacements such as weaves and braids, interloopings such as knits and crochets, and nonwovens such as felts, often next to other aspects like embellishments, fibre composition, and assembly methods (cf. e.g. Fashionary, 2020; Sarkar et al., 2023; Thompson, 2014). This organisation according to principal structural qualities reflects the consideration of textiles as goods whose design and production process is terminated that has been discussed in the first subsection.

A focus on finite, existing textiles is also characteristic of Emery's (1966) seminal work *The Primary Structures of Fabrics: An Illustrated Classification*, as Bühler (1973) notes in his introduction to Seiler-Baldinger's *Systematik der Textilen Techniken*. Consequently, he states that Emery's structural focus leads to a description of weaves and braids within the same group, despite the fact that both require different production processes and tools. Seiler-Baldinger's work, by contrast, categorises textile techniques according to their production process and the complexity of the tools involved, which may lead to repetitions of identical structures across several categories. While both approaches can be considered complementary (Balfet & Desrosiers, 1988; Desrosiers, 2010), Seiler-Baldinger's focus on making processes is nevertheless of particular relevance for textile design as a hands-on research discipline with its agency of influencing the precise composition of a textile by determining its production process.

A comparable making-based approach underlies Speiser's (2024) *An Annotated Classification of Textile Techniques*. It can further be found in Collingwood's (2023) *The Techniques of Rug Weaving*, originally published in 1968. Though primarily focused on weaving, Collingwood highlights its relationships with other textile techniques by illustrating a broad spectrum of possibilities to arrange threads in rugs, and does so from a practitioner's perspective. He notably describes a technique he terms *weft looping*, where weft threads are pulled up from the surface of the weave to form loops, which are subsequently interlooped with adjacent pulled-up weft loops. This generates a textile that

is simultaneously interlaced and interlooped. By presenting several variations of this technique (Figure 1), Collingwood constructs a relationship between weaves and knits instead of distinguishing them based on their structural differences, thus blurring their categorical boundary. In reference to Figure 1b, he notes that if “the warp and the plain weft were extracted, a piece of normal weft knitting would be left” (p. 216). Additionally, his statement that “a study of knitting methods might lead to the discovery of other variations of this technique” (p. 218) can be interpreted as an encouragement for textile practitioners to explore such shifts of habitual segmentations further.

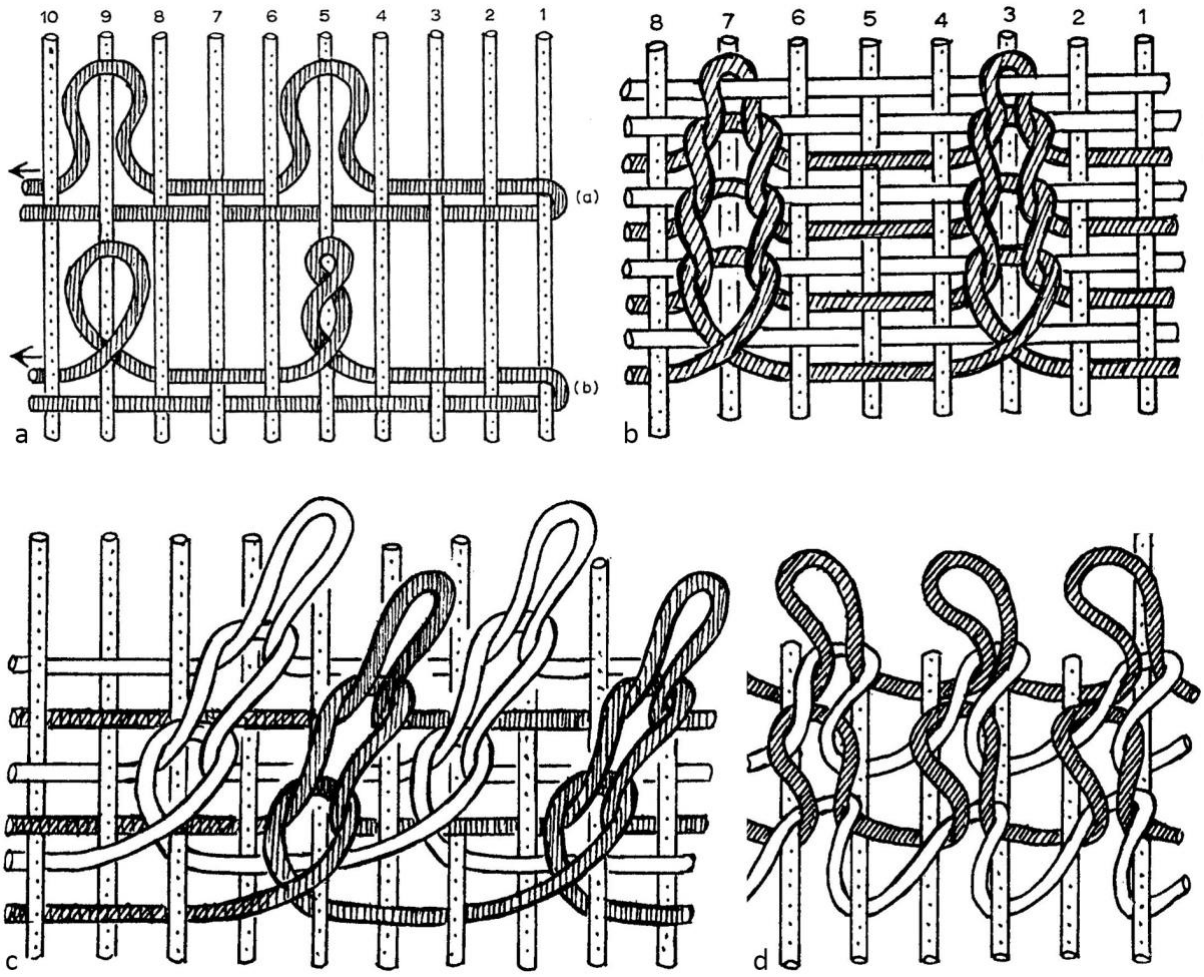


Figure 1. Variations of the Weft Looping technique described by Collingwood. a: Pulled-up loops. b-d: Variations of chained weft loops. Source: Collingwood (2023). The techniques of rug weaving. Echo Point Books & Media, LLC. Figs. 153, 158, 160, 161.

A different approach at renegotiating boundaries between textile categories is Harris' (2008) proposal for the term of "cloth-type materials" (p. 224) in the field of textile archaeology. She argues that the habitual separate consideration of human-made textiles and animal skins in this field can obscure the fact that both materials share flexible, shapeable qualities allowing them to be used for clothing or other forms of protective coverings. A terminological change towards cloth-type materials lays the focus on these qualities and permits to consider both types simultaneously, alongside other materials like "certain types of plastics, thin metal or large leaves" (p. 234). This permits insights into the connections between such materials and their simultaneous, interrelated use during human history. However, Harris underlines that she does not propose to "establish a rigid

taxonomy of cloth-type materials or material culture, but [that] exploring these levels of classification can be enlightening" (p. 233). A similar potential-focused approach emphasising similarities instead of differences and allowing for shifts of perspective is proposed in this paper.

A common characteristic of most of the classifications and categorisations presented here is their focus on textiles and techniques that already exist, without necessarily including how these could be developed further in the future. While this consideration may be beyond the scope of such works, an additional lens for approaching such classifications and categorisations from a practitioner's perspective, e.g. the one proposed in this paper, could create an openness to and inspiration for future developments of alternative textiles and techniques.

3. The thread arrangement as a lens for mapping the in-between

I propose the term "thread arrangement" as a conceptual lens and extension to conventional textile classifications and categorisations to account for the existence of hybrids between interlacements, interloopings and nonwovens. First, I provide a general overview of the research context the concept originates in, alongside methodological considerations on its development. In a second step, I present my use of this concept in my doctoral research, before presenting how it could be used in a wider context in the last subsection.

3.1. Research context and methodology

I used the term "thread arrangement" in my doctoral thesis while reflecting on my practice-based research in writing (Drews, forthcoming). In this research, I explored ways of arranging threads that permit the construction of complex three-dimensional textile artefacts such as garments without requiring any post-loom operations like cutting or sewing. In addition, I created textiles in which woven and knitted sections are seamlessly joined (Drews et al., 2023; Rosato et al., 2023).

In comparison to woven form construction based on the physiochemical qualities of the threads, e.g. through the use of shrinking yarns, my focus on alternative structural setups of the textile remains a relatively underexplored field of research (Albers, 2017). It therefore required the development of an inclusive vocabulary that could describe transgressions of the limitations of habitual definitions of textiles. However, the term "thread arrangement" was used rather intuitively in the thesis and not contextualised or reflected on further. In addition, the hybrids of knitted and woven structures resulting from my research were identified as potentially challenging to established classification systems, but have not yet been confronted with such systems in more detail.

Methodologically, the development of the term follows an inductive trajectory, using my own practice and writings about it as starting points for reflection and theorisation. By confronting this practice with existing frameworks of classification and identifying similar cases, I developed a conceptual and terminological lens based on these observations that is presented here. The term is thus informed by different levels of reflection happening at different points of the research: my own practical research on the loom, my written descriptions of this research, and the reflection on terminology developed during this process. Borrowing Schön's (2016) terms, this concept can therefore be considered the fruit both of a *reflection-in-action*, which happened during the weaving practice and, to some extent, while writing about it, and a *reflection-on-action* happening both during the writing process and my later meta-reflections presented here.

3.2. Concept development

In my doctoral thesis, I use the term "thread arrangement" as a conceptual extension to the word "textile" to underline two central aspects: Firstly, the textile's human (and thus designed) origin, and secondly, the potential versatility of its structural setup. The first aspect is expressed in my general definition of textiles:

> The term textile as used in this thesis refers to any human-made arrangement of threads and/or fibres. It is the outcome of a deliberate design process that defines its specific structural and physiochemical setup [...]. (Drews, forthcoming, p. 28)

The second aspect is expressed in the description of my research focus and is accompanied by a conceptual drawing (Figure 2):

> In this research, I focus on threads and their potential to be arranged in an infinite spectrum of ways ranging from a straight line, to differently sized loops and knots, to combinations of all of these variants [...]. (Drews, forthcoming, p. 82)

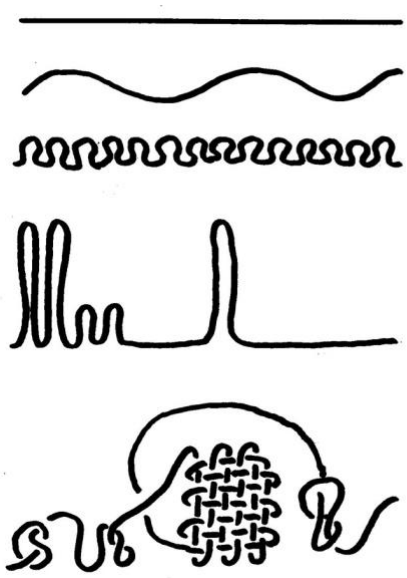


Figure 2. Conceptual drawing of different variations of thread arrangements. Source: Drews (forthcoming), p. 82.

At a later point, and in relation to the specific outcomes of my practice-based research, I sharpen my definition of textiles as follows, while integrating the aspect of textile constructions that deviate from habitual ones:

> A textile is a human-made thread arrangement that could be designed in many other shapes and forms than e.g. that of a planar, rectangular surface, and without relying on operations external to the production of the textile itself. (Drews, forthcoming, p. 281)

While the ideas of threads and their arrangement are central to all these considerations, the term "thread arrangement" is not further reflected upon in the thesis and was chosen rather intuitively. It is informed by a mindset considering textiles as designed material systems composed of threads, which are themselves composed of fibres, an organisation that can be traced down to the molecular level of the raw materials involved (cf. also Chen & Hearle, 2010; Fratzl & Schäffner, 2021; Tandler,

2016). At the same time, the specific approach of the research of building the final form directly from the threads omits the traditional gap between textile and 3D-form construction. No fabric in a traditional sense is produced here, but rather an integrated textile-form (McQuillan, 2020), in which the structural setup and final three-dimensional form are designed and produced simultaneously. More established terminology such as "textile structure", "textile construction", or "fabric architecture" may not have been descriptive enough of these circumstances, as they imply a regard on the textile *as is*, without taking into account its process of becoming (cf. the critique on Emery's classification described above). By contrast, the term "thread arrangement" distinctly mentions the textile designer's primary material, threads, as well as their often hands-on activity of arranging them to create a textile. It is used as a term to qualify the finished textile artefact and its design as an integrated process.

The concept proved further useful as a meta-category encompassing both woven and knitted textiles, as well as hybrids between them. While weaving was taken as a starting point to explore form making, I also developed possibilities for seamless transitions between knits and weaves within the same textile, which I describe in more detail elsewhere (Drews, 2019, forthcoming; Drews et al., 2023; Rosato et al., 2023). In these examples, threads begin their journey through the textile as straight lines or interlacements, before being transformed into interlooped knits and vice versa. This creates a textile surface seamlessly joining more rigid woven sections and more flexible knitted ones and provides opportunities for the development of seamless textile artefacts where a zoned flexibility is desired. I have demonstrated this principle through the creation of a trouser prototype, where the knitted zone can adapt to variations of the wearer's waist circumference (Figure 3a), and in a prototype for a saddle cover developed by students under my guidance, whose knitted zone provides enough elasticity to be stretched over the saddle (Figure 3b). In these examples individual sections can be clearly identified as woven or knitted, but the textile as a whole belongs to both categories.

This practical exploration of weave/knit hybrids, alongside the study of related textiles merging different types of thread arrangements such as Collingwood's, revealed a significant gap: while these structures exist as physical realities, they cannot be properly integrated into classification systems that distinguish textiles on the base of their overall structural composition (e.g. weaves or knits). In the thesis, I have therefore identified such textiles as "a starting point for a criticism of textile classification systems that solely distinguish between *principal* thread arrangements (Drews, forthcoming, p. 282). This notion will be further elaborated upon in the following.



Figure 3. Seamless weave/knit hybrids. a: Seamlessly woven trouser prototype developed by the author, featuring a frontal knitted section contributed by Dr. Ludovica Rosato (Bologna University), which serves as a closing mechanism. b: Saddle cover prototype seamlessly combining woven and knitted sections, developed by students Max Cassagnes, Coline Girard and Kenza Ka under the author's guidance.

3.3. The thread arrangement as a terminological and conceptual extension to textile classification

To increase the visibility and relevance of "cases that do not fit" (Bowker & Star, 1999, p. 9) into habitual classification systems, and in analogy to Harris' (2008) concept of cloth-type materials, I propose the meta-category of the thread arrangement as a lens to read textile classifications and to

emphasise the porosity of their internal segmentations. This concept is not intended to replace existing classifications, but to act as a potential- and practice-focused extension to them that can inspire further research into subjects such as textile hybridity. It aims to allow for the inclusion of future insights and developments that may shift existing boundaries, e.g. those between the design of textiles and that of final textile artefacts, or between knits and weaves, and thus to help in maintaining the relevance of existing categorisations and classifications.

The term "thread arrangement" addresses the principal aspects worked out in the theoretical framework above. While describing the textile in its finished state, it simultaneously implies a previous act of arranging. This duality is essential to acknowledging the creative agency of the textile designer, whose profound knowledge of materials, tools, and possibilities for construction allows them to arrange threads in myriad ways. Through the inclusion of this perspective of a creative tinkerer pushing the boundaries of what is considered normal (Barati & Karana, 2019; Glăveanu, 2012; Lehmann, 2012), the divisive setup of many textile classifications can seem less rigid and is more aligned with the reality of the hands-on experience of organising threads into textiles in any way the designer sees fit.

In addition, this broader, more flexible meta-category encompassing any type of structural composition provides a space for the inclusion of the principles of ambiguity, adaptability to change, and sensitivity to exclusion proposed by Bowker and Star (1999). It permits to qualify textiles as designed systems of arranged threads without immediately labelling them as weaves, knits, or nonwovens. And lastly, like Harris' (2008) cloth-type materials, the lens of the thread arrangement permits a focus on similarities between textiles while toning down their differences emphasised in many habitual classifications.

The lens of the thread arrangement could therefore serve as a reminder for textile practitioners that most textile classifications and categorisations only describe previously existing textiles, but that this does not mean that their threads could not potentially be arranged differently. It could serve as a tool to deconstruct existing boundaries and reassure textile designers of their creative agency, notably by triggering questions such as: Why did someone decide to arrange the threads in this way? Could they have been arranged differently? What kinds of tools would be necessary to do so? Through such interrogations, existing textile classifications could become invitations to creatively explore new types of thread arrangements.

In summary, the conceptual lens of the thread arrangement arose from the necessity of qualifying textiles transgressing categorisations into weaves, knits and nonwovens in a more nuanced way. It can therefore be seen as an inclusive expansion of the vocabulary of textile design that emphasises the discipline's creative agency. It is further intended as a conceptual tool to accompany creative tinkering and to facilitate further research into the potentials arising from the textile's structural organisation.

4. Reflections: Potential expansions and limitations

In the future, the thread arrangement framework could be enriched by further examples of hybrid textiles, as well as analyses of expressive, functional, and conceptual properties shared between weaves, knits, nonwovens, and further thread arrangements. These could e.g. include the elasticity

of weaves in the bias direction (cf. e.g. Penava et al., 2020) to deconstruct the dichotomy of the rigid weave and the stretchable knit, or the terminological overlap between knits and weaves in the terms "weft knit" and "warp knit" (cf. e.g. Drews, forthcoming).

A critical reflection on this lens reveals a potential limitation, namely its focus on the thread rather than the fibre, which stems from the particular context of the research it was developed in. I have opted for "thread arrangement" to emphasise the rootedness of the concept within the discipline of textile design, whose practitioners primarily work with threads and only more exceptionally compose these themselves from fibres. However, in different research contexts, e.g. such more preoccupied with the field of nonwovens, the term "fibre arrangement" may be more appropriate. In addition, overlaps between the scale of the fibre and that of the thread may occur, as expressed e.g. in the technique of fulling. In this wet finishing for woollen weaves, the textile is mechanically treated to induce fibre interlocking on and between the interlaced threads, thus creating a felt-like material with a woven base structure (Kilburn Scott, 1931; Paardekooper, 2005; Soriga, 2017).

Consequently, practitioners may have to operationalise the proposed lens differently, depending on their individual research context. Rather than as a fixed term, the concept should therefore be seen as an encouragement for textile designers to develop precise terminology that is appropriate to their research and the shifts of perspective it may bring about and to contribute this terminology as a base for discussion to the field of textile design research. First and foremost, it is proposed as a space for acknowledging textile designers' creative agency and the fact that textiles could be designed differently, and for inspiring and fostering future research into textile hybrids and other creative shifts of habitual boundaries.

5. Conclusion

This paper has argued for the necessity of a more fluid, inclusive, and potential-focused vocabulary in the field of textile design to qualify hybrid textile structures. By introducing the "thread arrangement" as a conceptual lens, the research moves beyond the potentially divisive logic of established classification systems that sometimes fail to encompass the complexity of such textiles. Drawing from textile design, creativity, and classification theory, as well as insights from textile archaeology, this framework presents textiles not as fixed categories, but as outcomes and expressions of a textile designer’s research process and creative agency.

In addition to a reflective analysis of the emergence of the concept in my doctoral research and an interdisciplinary conceptual framework, examples of textiles and techniques have demonstrated that structural hybridity exists and can be a fertile ground for critical enquiry into existing classifications. Conceptualising textiles as thread arrangements acknowledges that every fabric is a human-made system that could have been designed differently. This shift in perspective empowers textile designers to act as arrangers of potentials, fostering a research culture that values the creative transgression of norms as a driver for explorations into undiscovered structural potentials.

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